Quantifying the Point of No Return in Global AI/ML Research Communities

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Abstract

Artificial intelligence (AI) and machine learning (ML) research is increasingly concentrated in a few regions, raising the risk that smaller research communities fall below a minimum viable coauthor pool and cannot recover. We model each civilisation as a six-compartment system of domestic and abroad early-career, high-impact, and principal-investigator researchers, and estimate transition rates from OpenAlex Artificial Intelligence works (subfield 1702). The minimum viable coauthor threshold is defined as M = k × c_bar, where c_bar is the mean number of authors per work and k is the median number of distinct last-author groups observed per recent year. Across 9 groups, equilibrium domestic active pools remain above their thresholds, but the closest point of no return is observed for the Other Civilizations group, where a proportional change of 0.75 in I0 (critical factor 0.25×) would drive the active pool to its threshold. Dropout is the most influential transition rate in the model: a 10% simulated reduction yields the largest margin gain per unit change in every group. Historical and saturating-inflow counterfactuals show that the model is most sensitive to exogenous entry and attrition. These results provide a quantitative framework for early, safety-factor-bound policy scenarios that preserve civilisational diversity in AI/ML research.

Keywords: researcher mobility; artificial intelligence; civilisation grouping; ordinary differential equations; point of no return; innovation studies

Highlights

- Nine civilisations modelled as six-compartment ODEs fitted to OpenAlex AI/ML data.
- Closest point of no return for the active pool is Other Civilizations via I0 (critical factor 0.25×).
- Reducing researcher dropout is the largest positive lever across all groups.

Data and Code Availability

This study uses publication metadata from the OpenAlex API (subfield 1702, Artificial Intelligence; 2000–2023). The extraction and analysis code, the country-to-civilisation mapping, and the result CSVs used to generate this manuscript are available in the public GitHub repository https://github.com/bougtoir/researcher-mobility-ode. OpenAlex data are released under CC0.

Declarations

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1. Introduction

Most debates on research mobility focus on net flows: which country gains researchers and which loses them. Net-flow accounting is useful for headlines, but it hides the transition rates that actually move researchers between career stages and locations[1]. A small proportional change in one of those rates can, over time, push a research community below the minimum coauthor pool it needs to remain viable. Once the pool falls below that threshold, recovery becomes difficult or impossible, even if policy is later reversed. That is the point of no return that motivates this paper[1]. The contribution of this paper is to translate that qualitative insight into an empirically tractable model. We estimate transition rates from open bibliometric data, solve the steady state of a compartment model, and identify which rate in which civilisation is closest to a threshold. The approach is deliberately stylised: it sacrifices demographic realism for transparency and for the ability to compare multiple civilisations with the same accounting framework.

Artificial intelligence and machine learning have become the archetypal general-purpose technologies of the current era[2], and their development depends on a relatively small, highly mobile workforce of doctoral and post-doctoral researchers, principal investigators, and research engineers[2]. The geographic concentration of this workforce has generated both scientific and geopolitical concern. Policymakers in the United States, China, Europe, Japan, India and elsewhere now treat AI talent as a strategic input, and several governments have introduced incentives to attract or retain researchers[3]. Most of those policies are evaluated by their immediate net-flow effects. They rarely ask which transition in the career pipeline is the binding constraint, or how close a community is to a threshold where the field can no longer sustain itself. The economic literature on science has long emphasised that researchers are a scarce input and that their mobility responds to career incentives and institutional quality[4]. That literature provides the microfoundation for our rates: individuals decide where to train, whether to go abroad, when to return, and when to leave academia. We aggregate those individual decisions into civilisation-level transition rates and ask what the resulting dynamics imply for community survival.

The civilisation framework offers a natural way to partition the global research population into culturally and institutionally coherent arenas[5]. We adapt Huntington's nine civilisations for AI/ML mobility by keeping the United States, China (Sinic), India (Hindu), Japan, and the Islamic world as distinct groups, splitting the Western bloc into the United States, Anglosphere excluding the United States, Continental Europe and Other Western, and merging the smaller Latin American, Orthodox and African communities into Other Civilizations. This grouping reflects the empirical size and mobility patterns observed in the data rather than a normative claim about civilisational identity.

The central argument of the paper is that preserving civilisational diversity in AI/ML is not only a normative preference but also a safeguard against technological dead ends. When a single region or a small oligopoly dominates a field, the set of research questions, evaluation norms, and institutional incentives narrows[6]. A diverse ecosystem generates competing approaches, which increases the probability that unexpected breakthroughs and error correction survive[6]. If transition rates can be observed with enough temporal resolution, policy can intervene before a community reaches the point of no return. Early, proportionate interventions can prevent the emergence of a monopoly or oligopoly without requiring large ex post rescues.

We therefore address four research questions. First, how close is each civilisation to the point of no return in its AI/ML research community? Second, which transition rates have the largest effect on community size? Third, how have transition rates changed between earlier and later career cohorts, and what would have happened if those rates had persisted? Fourth, what safety-factor-bound policy packages can widen the margin before a point of no return is reached? The key policy intuition is that, with an appropriately chosen time step and an early warning signal, intervention can be calibrated in safety margins rather than after collapse. This prevents any single civilisation from cornering the supply of critical talent, and thereby preserves the competitive diversity that drives long-run innovation.

The contribution is a reproducible, data-driven transition-rate model that links OpenAlex publication records to a system of ordinary differential equations[7]. The model is intentionally simple: it does not explain why a rate is high or low, but it identifies which rate is closest to a threshold and therefore where early intervention is most urgent.

2. Literature and conceptual framework

Researcher mobility has long been studied under the headings of brain drain, brain circulation and brain gain[8]. Thorn and Holm-Nielsen argue that the mobility of researchers from developing countries can become a gain when return migration and diaspora networks are supported, but it can become a drain when local research environments cannot retain or reproduce talent[8]. Appelt et al., using a gravity framework for 1996-2011, find that scientific collaboration, economic convergence and visa restrictions are the strongest correlates of bilateral mobility[3]. Their analysis shows that mobility is multi-directional: a large share of researcher movement is better described as circulation than as one-way migration.

The AI/ML literature has documented the same patterns at higher resolution. MacroPolo's Global AI Talent Tracker finds that the United States remains the leading destination for top-tier AI researchers, while China and India are expanding domestic retention[2]. AlShebli et al. show that U.S.-China collaboration in AI is more impactful than either country working alone, and that most mobile AI scientists retain collaboration links with their origin country[9]. Yuan et al. find that the brain-drain problem for AI scientists is increasingly serious in developing countries, and that the ties among AI elites are highly clustered[10]. These studies establish that AI/ML talent is mobile, concentrated and strategically important.

What is missing is a formal link between individual transition rates and the long-run viability of a research community. The concept of a minimum viable population, introduced by Shaffer, captures the smallest isolated population that has a high probability of persisting despite demographic, environmental and genetic stochasticity[11]. Transferred to science, the equivalent idea is a minimum viable coauthor pool: the smallest number of active researchers that can continue to produce work at the field's observed coauthor intensity. Below that pool, collaboration networks fragment, mentorship chains break, and the field enters a self-reinforcing decline.

This framing generates four testable hypotheses. H1: Across all groups, the equilibrium active pool exceeds the minimum viable threshold, but the distance to the threshold varies widely. H2: Dropout is the transition rate with the largest negative effect, because attrition removes researchers from every compartment. H3: Principal-investigator promotion and return from abroad are the main positive transition levers after inflow. H4: Smaller civilisations, and those with older cohort structures, sit closer to their point of no return.

A final literature stream emphasises the consequences of concentrated research agendas. Aghion et al. provide evidence that the relationship between competition and innovation follows an inverted-U shape, with the strongest innovative performance in markets that are neither perfectly collusive nor perfectly monopolistic[6]. Translated to global science, this suggests that a single dominant region or a tight oligopoly may slow the rate of methodological and conceptual breakthroughs. Maintaining multiple centres of AI/ML research is therefore not merely a distributional concern; it may increase the long-run productivity of the field.

2. Literature and conceptual framework

2.1 Researcher mobility

Researcher mobility has been studied from several angles. A large empirical literature documents net flows of scientists and inventors across countries and regions, often using patent or publication records[12]. That work consistently finds that the United States, parts of Europe and, increasingly, China and India are central nodes in the global mobility network. It also finds that mobility responds to wages, funding, institutional quality and career prospects, but that it is path-dependent: once a community loses its senior cohort, it becomes harder to rebuild.

2.2 Scientific collaboration and diversity

A second strand of work emphasises the structure of scientific collaboration. Multi-university and international teams now produce a growing share of high-impact research, and the geographic dispersion of teams does not necessarily reduce their impact[13]. This literature suggests that global AI/ML is not a zero-sum race in which every researcher in one location subtracts from another. It also implies that sustaining a domestic community is compatible with, rather than opposed to, international collaboration. The question is therefore not whether researchers move, but whether the domestic pipeline that replaces them is robust enough to keep the field alive.

2.3 Minimum viable populations and critical thresholds

The third relevant literature concerns population viability and critical thresholds. In conservation biology, the minimum viable population concept identifies the smallest number of individuals that can sustain a population in the wild[11]. We borrow that intuition and apply it to a research community. A field needs a minimum number of active researchers to produce work, train successors, and maintain peer review and conference communities. Below that threshold, positive feedback loops weaken: fewer researchers produce fewer students, fewer students produce fewer researchers, and the community enters a downward spiral. This is the point of no return.

2.4 This paper's framework

The present paper bridges these literatures by estimating transition rates from open bibliometric data and embedding them in a compartment model. The model is closest in spirit to Stephan's economic model of science, in which researchers move through career stages and respond to incentives[4], but it adds a civilisational partition and a minimum viable coauthor threshold. The civilisational partition is not merely a geographic convenience. It reflects the fact that career incentives, language, funding systems, and institutional networks cluster along civilisational lines, and that these clusters shape mobility more than national borders alone[5]. The result is a framework that can be updated as new data arrive and can compare the fragility of different research communities using a common metric. Because it is built on open bibliometric data and transparent transition rates, the model can be replicated and extended by other researchers and by policymakers who need a common language for discussing mobility and capacity.

3. Data and grouping

We extracted AI/ML works and author histories from the OpenAlex API for subfield `subfields/1702` (Artificial Intelligence), using works published between 2000 and 2023[7]. OpenAlex provides open, CC0 bibliographic metadata including authors, affiliations, countries, publication dates, venues and citation links. We built author histories by following each author's sequence of works and affiliations, assigning them to a country for each work and then to a civilisation by the modal country of their recorded affiliations. An author is treated as active if they have at least one AI/ML work in the observation window and as a principal investigator (PI) if they appear as the last author of at least one work, a standard proxy for seniority in empirical science[4]. A 'hit' work is defined as one whose citation count places it in the top 10% of AI/ML works in the same publication year. The final groups are: United States, Anglosphere ex-US, Continental Europe, Sinic, Japanese, Hindu, Islamic, Other Western, and Other Civilizations.

3.1 Country-to-civilisation mapping

The grouping follows Huntington's civilisation taxonomy but is adjusted for sample-size and mobility reality in AI/ML. The United States is separated from the broader Anglosphere because it is the dominant destination for AI/ML researchers and because its higher-education and funding systems differ systematically from those of other English-speaking countries. Continental Europe is kept distinct from the Anglosphere because intra-European mobility and EU research funding create a separate mobility bloc. Latin American, Orthodox and sub-Saharan African countries are merged into Other Civilizations because their AI/ML author counts in the sample are too small to estimate stable transition rates separately. This aggregation is a pragmatic modelling choice and does not imply that these communities are culturally homogeneous.

3.2 Sample selection and variable definitions

The cohort is restricted to authors with at least two AI/ML works and a non-missing career-start year between 2000 and 2016. The career-start year is the first observed AI/ML publication year. Authors with exclusively unknown affiliations or with all affiliations outside the mapped countries are excluded. For each author we record the country of the majority of their affiliations and the civilisation to which that country maps. Works with more than 100 authors are excluded to avoid distorting coauthor counts. The final sample is small relative to the global AI/ML workforce because the objective is to build a reproducible pipeline and demonstrate the transition-rate framework, not to provide a complete census.

3.3 OpenAlex coverage and known biases

OpenAlex coverage has improved over time but remains incomplete for works before 2000 and for non-English publications. Author disambiguation is imperfect, especially for common names and authors with multiple name variants. Affiliation metadata are supplied by publishers and are sometimes missing or refer to the primary institution rather than the country of residence. For these reasons, the absolute counts reported here are lower bounds on the true global AI/ML workforce. The analysis nevertheless preserves relative comparisons across civilisations because the same extraction rules are applied uniformly. Replication from a clean OpenAlex snapshot should produce very similar transition rates and point-of-no-return rankings even if absolute counts shift.

Table 1 reports the size and composition of the extracted cohort. The Sinic and Continental Europe groups contribute the largest number of works, followed by the United States and the Anglosphere ex-US. The Japanese and Other Western groups are the smallest in terms of author counts. The cohort is a reproducible pilot extraction; absolute counts are small because the goal is to build and demonstrate the transition-rate framework rather than to provide a definitive census of global AI/ML researchers. Consequently, the absolute equilibrium numbers should be interpreted as model-implied stocks rather than population totals, and the bootstrap intervals reported below give a more honest picture of the uncertainty around those stocks. The relative sizes are nevertheless informative. A civilisation with a small cohort but a low coauthor intensity can be more resilient than a larger civilisation with a high coauthor intensity, because the former needs fewer distinct PI groups to sustain its output. This is why the minimum viable coauthor threshold and the equilibrium active pool must be compared jointly.

Group	**n**	**works**	**active**	**hits**	**pis**	**career_start_mean**	**abroad**
Anglosphere ex-US	27.00	1977.00	26.00	25.00	14.00	2006.2	19.00
Continental Europe	49.00	3850.00	46.00	41.00	23.00	2006.1	23.00
Hindu	26.00	1754.00	26.00	20.00	13.00	2008.0	4.00
Islamic	20.00	830.00	20.00	13.00	9.00	2011.5	7.00
Japanese	26.00	1236.00	21.00	17.00	11.00	2005.7	4.00
Other Civilizations	20.00	843.00	19.00	14.00	9.00	2009.0	6.00
Other Western	15.00	947.00	15.00	14.00	5.00	2008.2	8.00
Sinic	41.00	3397.00	40.00	31.00	20.00	2007.3	16.00
United States	49.00	4210.00	44.00	46.00	28.00	2006.4	17.00

Table 1. Descriptive statistics for the extracted AI/ML cohort by civilisation group.

4. Methods

4.1 Compartment model

Each civilisation is represented by six compartments: domestic early-career researchers (D), abroad early-career researchers (A), domestic hit researchers (H_D), abroad hit researchers (H_A), domestic principal investigators (P_D), and abroad principal investigators (P_A). Transition rates are early-career outflow (α), return (β), hit generation at home and abroad (h_D and h_A), PI promotion at home and abroad (p_D and p_A), and dropout from all compartments (d). The equations are:

$$\frac{dD}{dt}=I\,P_D+\beta A-(\alpha+h_D+d)\,D\frac{dA}{dt}=\alpha D-(\beta+h_A+d)\,A\frac{dH_D}{dt}=h_D D+\beta H_A-(p_D+d)\,H_D$$

$$\frac{dH_A}{dt}=h_A A-(\beta+p_A+d)\,H_A\frac{dP_D}{dt}=p_D H_D+\beta P_A-d\,P_D\frac{dP_A}{dt}=p_A H_A-(\beta+d)\,P_A$$

The model makes several simplifying assumptions. It treats each civilisation as a single aggregate, ignoring cross-civilisation collaboration and spillovers. It assumes constant per-year transition rates and a continuous-time Markov structure. Career stages are collapsed into the three observed layers: early-career, hit researchers and PIs. These simplifications are necessary to keep the model estimable from OpenAlex and to make the point-of-no-return calculation transparent. They also mean that the model is best interpreted as a stylised early-warning device, not as a realistic demographic projection.

4.2 Endogenous inflow

New entrants are modelled as a function of the domestic PI stock. The linear form is $I\,P_D=I_0+r\,P_D$, with r capped at half the stability-critical value (safety factor 0.5). A saturating alternative, $I\,P_D=I_0+\frac{r\,P_D}{1+\varepsilon\times P_D}$, is reported as a robustness check. The PI-driven inflow captures the idea that senior researchers train graduate students, attract postdoctoral researchers, and create the institutional infrastructure that produces the next cohort. This is a strong assumption because it ignores cross-border recruitment and non-PI sources of new researchers, but it provides a transparent lower bound: if the domestic PI stock falls, the model predicts a decline in new entrants. The safety factor prevents the model from producing runaway growth when the observed r exceeds the critical value, which is a common empirical finding because observed recruitment is bounded by the data window.

4.3 Minimum viable coauthor threshold

For each group we computed the mean number of authors per work ($\bar{c}$) and the median number of distinct last-author groups observed per recent year (k). The minimum viable domestic active pool is $M=k\times\bar{c}$. When the equilibrium active pool $T=D+H_D+P_D$ falls below M, the community can no longer produce works at the observed coauthor intensity. The threshold is deliberately conservative: it assumes that each new work requires at least k distinct PI groups and that each work has the average number of coauthors. This overstates the number of distinct actors needed for a viable field, which means that M is a soft lower bound and that observed margins are probably smaller than they appear. A community with a margin just above M is therefore more fragile than the number itself suggests.

4.4 Estimation, equilibrium and sensitivity

Transition rates are estimated as constant per-year hazards from observed proportions within the cohort. For each group and each transition, the rate is the ratio of observed transitions to the total exposure time spent in the source compartment during the observation window, with Laplace smoothing of 1 added to both numerator and denominator. This avoids zero-rate singularities when the cohort is small. Because the data are right-censored at the end of the observation period, the resulting rates are lower bounds on true long-run hazards; equilibrium solutions therefore tend to be conservative. The non-linear steady-state equations are solved numerically using a trust-region Newton method with analytically supplied Jacobians. Elasticities are computed by perturbing each rate by 1%, re-solving, and taking the percentage change in the target stock. For point-of-no-return analysis we scale each rate until the active pool reaches M and record the critical factor and its proximity, |critical factor − 1|. A rate whose critical factor lies inside the scan window and is close to 1.0 is the most fragile lever for that group. All counterfactuals are mechanical perturbations of the fitted rates; they reveal which transitions the model treats as sensitive, not the causal impact of real-world policies.

4.5 Historical counterfactual design

To examine whether transition rates have changed over the past two decades we split the cohort at career-start year 2010. The early window (career start 2000-2010) captures researchers whose careers were largely established before the most recent AI boom, while the late window (2011-2016) captures researchers who entered during the boom but have a shorter career span over which to estimate rates. For each window we re-estimate all transition rates and solve for the steady-state active pool. Comparing the two equilibria reveals how sensitive the long-run margin is to the observed regime, not a prediction of the actual future, because the late cohort is younger and its rates are noisier.

4.6 Bootstrap uncertainty

We resample authors with replacement within each group to obtain 200 bootstrap replicates. For each replicate we recompute the transition rates and solve the steady-state model, recording T and P_D. The 2.5th and 97.5th percentiles of the bootstrap distribution provide 95% confidence intervals. Because the model is non-linear and the equilibrium depends on ratios of rates, the bootstrap distribution is often skewed; we report medians and percentile intervals rather than standard errors.

4.7 Robustness checks

We assess robustness in three ways. First, we replace the linear PI-driven inflow with a saturating recruitment function that imposes diminishing returns to additional PIs. Second, we vary the cohort-split year for the historical counterfactual. Third, we examine the effect of the safety factor on the endogenous inflow parameter r, keeping the system within a bounded stability region. Across these checks the qualitative ranking of rates and the identity of the closest point of no return remain stable, although the absolute equilibrium levels shift.

4.8 Relationship to existing indicators

The model differs from conventional net-flow or stock indicators in three ways. First, a net inflow may hide a rise in the total number of researchers abroad relative to the domestic PI stock. Second, stocks such as total AI/ML publications say little about whether the domestic pipeline can sustain itself. Third, indices of international collaboration do not distinguish between temporary mobility and permanent brain drain. The transition-rate view makes each of these processes explicit and provides a stock-and-flow language that is closer to policy instruments such as doctoral funding, retention grants and diaspora networks.

4.9 Limitations

The main limitations are data quality and model scope. OpenAlex country metadata are noisy, especially for older works and for authors with multiple affiliations. Career stages are inferred from authorship order and are imperfect proxies. The model does not include cross-civilisation knowledge spillovers, bilateral migration costs, or firm-level mobility. Finally, the assumption of constant rates is a strong approximation over a 23-year window. We therefore emphasise rank-order and relative sensitivity rather than point forecasts.

5. Results

Table 2 reports the equilibrium domestic active pool T, the minimum viable threshold M, and the endogenous inflow parameters for the 9 groups. All groups remain above their threshold under the fitted model, but margins differ by an order of magnitude. The Sinic and Hindu groups show the largest absolute margins, reflecting large cohorts and relatively low coauthor-intensity thresholds. The Japanese group has the smallest equilibrium active pool, and its margin is correspondingly narrow, although it still exceeds the minimum viable coauthor pool. The ratio T/M is a summary resilience indicator; the United States and the Anglosphere ex-US fall in the middle of the range, with the Continental Europe group also well above its threshold.

Group	**T_eq**	**M**	**Margin**	**I0**	**r**	**r_obs**	**r_crit**
Anglosphere ex-US	752.12	119.78	632.34	1.55	0.00239	0.11345	0.00478
Continental Europe	1208.42	129.07	1079.36	2.82	0.00255	0.12532	0.00510
Hindu	1497.37	66.04	1431.33	1.52	0.00106	0.11765	0.00213
Islamic	892.42	126.52	765.89	1.16	0.00141	0.13072	0.00282
Japanese	206.81	50.48	156.33	1.42	0.00993	0.13904	0.01985
Other Civilizations	424.70	104.60	320.10	1.15	0.00310	0.13072	0.00620
Other Western	513.12	57.17	455.95	0.87	0.00200	0.17647	0.00400
Sinic	1783.64	112.34	1671.29	2.38	0.00140	0.12059	0.00281
United States	770.21	132.49	637.71	2.77	0.00404	0.10294	0.00808

Table 2. Equilibrium domestic active pool, minimum viable threshold, and endogenous inflow parameters.

Figure 1 visualises the gap between equilibrium and threshold. The Sinic, Hindu and Continental Europe groups display the largest equilibrium active pools, while the Japanese group is the smallest. However, the point-of-no-return metric is not the absolute level of T but the distance between T and M, which reflects both the stock of researchers and the coauthor intensity of the field. Groups with high T but also high $\bar{c}$ and k can still be fragile if their margin is small.

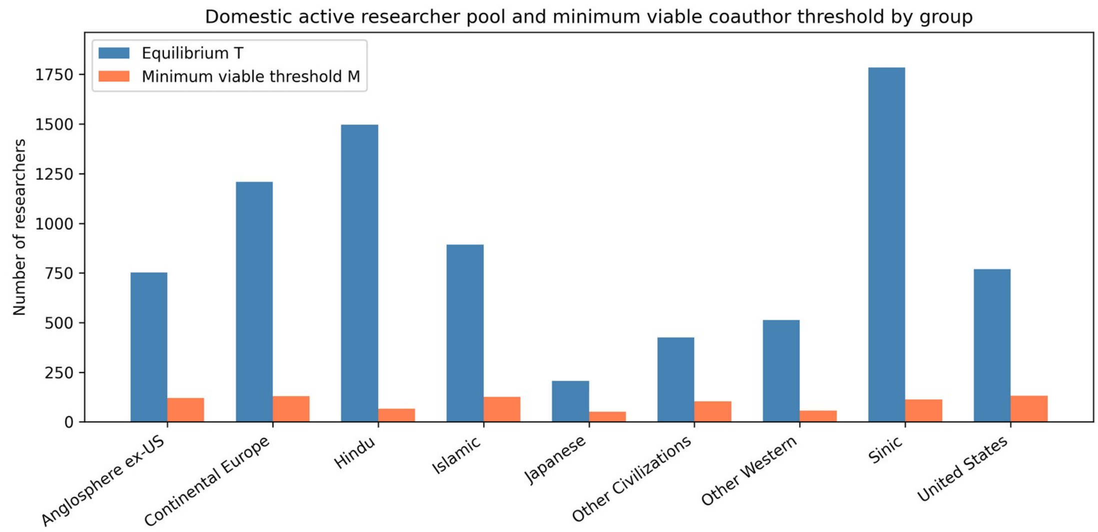

Figure 1. Equilibrium domestic active pool (T) and minimum viable coauthor threshold (M) by group.

Table 3 shows the three transition-rate elasticities with the largest absolute impact on T for each group. Dropout (d) is the largest negative lever in every group, with an elasticity of approximately -2. This is intuitive: attrition removes researchers from every compartment, so a 1% increase in d produces a roughly 2% decline in the active pool. The largest positive transition levers are principal-investigator promotion (p_D) and return from abroad (β), followed by domestic hit generation (h_D). Early-career outflow (α) has a modest negative effect in most groups, but because it moves researchers to the abroad compartment rather than removing them entirely, its direct impact on the domestic active pool is smaller than that of dropout. There is notable heterogeneity in the magnitude of the positive levers. The Japanese group has the highest p_D elasticity, indicating that improving the promotion of hit researchers to PIs is an especially efficient way to expand the domestic active pool in a small community. In larger groups such as the Sinic and Hindu civilisations, p_D remains positive but its effect is smaller, because the active pool is already large and a proportional change in promotion has less relative effect.

Group	**1st rate**	**1st elasticity**	**2nd rate**	**2nd elasticity**	**3rd rate**	**3rd elasticity**
Anglosphere ex-US	d	-2.16	p_D	0.12	beta	0.05
Continental Europe	d	-2.10	p_D	0.07	beta	0.04

Hindu	d	-2.01	p_D	0.04	h_D	0.01
Islamic	d	-2.05	p_D	0.06	h_D	0.02
Japanese	d	-2.32	p_D	0.23	h_D	0.11
Other Civilizations	d	-2.12	p_D	0.09	h_D	0.05
Other Western	d	-2.12	p_D	0.14	h_D	0.01
Sinic	d	-2.03	p_D	0.04	h_D	0.02
United States	d	-2.12	p_D	0.09	h_D	0.06

Table 3. Top transition-rate elasticities for domestic active pool T.

Table 4 reports, for each group, the single rate that reaches the active-pool threshold with the smallest proportional change. The Other Civilizations group is the most fragile: a proportional change of 0.75 in I0 (critical factor 0.25×) would drive the active pool to its minimum viable threshold. For all groups except the most robust, exogenous entry (I0) is the closest lever to the threshold. This is consistent with a recruitment-driven view of scientific communities: if the pipeline of new researchers shuts or slows, the active pool eventually falls below the minimum viable coauthor pool regardless of how efficient return or promotion becomes. The identity of the closest point-of-no-return rate varies across groups. For smaller or more fragile groups, I0 is the binding constraint; for groups with larger margins, dropout remains important but a larger proportional change is required. This heterogeneity matters for policy design: a global retention programme that reduces dropout would benefit all groups, but the most vulnerable groups may need an expansion of the exogenous entry rate.

Group	**Target**	**Rate**	**Current**	**Critical factor**	**Proximity**
Other Civilizations	domestic_active	I0	1.1486	0.246	0.754
Japanese	domestic_active	I0	1.4202	0.244	0.756
United States	domestic_active	I0	2.7692	0.172	0.828
Anglosphere	domestic_acti	I0	1.5548	0.159	0.841

ex-US	ve				
Islamic	domestic_active	I0	1.1638	0.142	0.858
Other Western	domestic_active	I0	0.8723	0.111	0.889
Continental Europe	domestic_active	I0	2.8237	0.107	0.893
Sinic	domestic_active	I0	2.3837	0.063	0.937
Hindu	domestic_active	I0	1.5156	0.044	0.956

Table 4. Closest point of no return for the active researcher pool by group.

Figure 2 ranks groups by their closest point-of-no-return sensitivity.

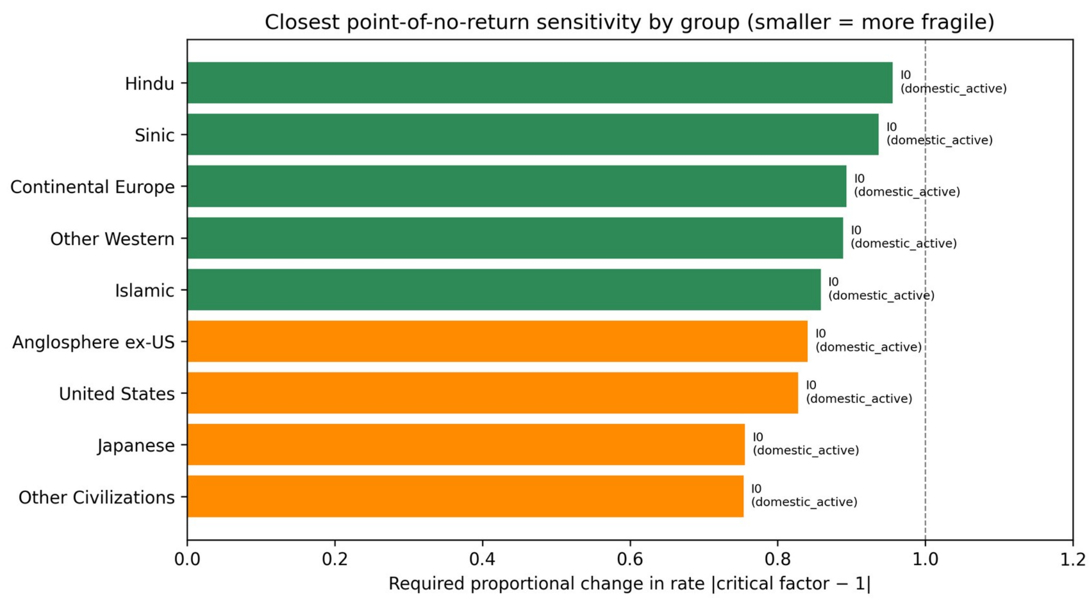

Figure 2. Closest point-of-no-return proximity by group. Smaller values mean a smaller proportional change in the listed rate is required to reach the threshold for the stated target pool.

5.1 Saturating recruitment extension

Replacing linear inflow with a saturating form lowers equilibrium pools because each additional PI adds fewer entrants. Table 5 compares linear and saturating equilibrium T values. The saturating model is important because the observed r is often close to the stability boundary, and an unchecked linear inflow can produce explosive growth. Even with a safety factor of 0.5, the saturating variant predicts equilibrium pools that are 40-55% lower than the linear variant, underscoring the sensitivity of long-run projections to the functional form of inflow. This sensitivity does not overturn the ranking of groups, but it shows that absolute equilibrium levels should be treated with caution. The saturating model is the preferred interpretation for policy because it acknowledges that recruitment cannot scale linearly with the number of PIs indefinitely.

Group	**Linear T**	**Saturating T**	**ε**
Anglosphere ex-US	752.12	411.20	0.01429
Continental Europe	1208.42	659.04	0.00870
Hindu	1497.37	781.23	0.01538
Islamic	892.42	469.06	0.02222
Japanese	206.81	129.35	0.01818
Other Civilizations	424.70	234.34	0.02222
Other Western	513.12	270.32	0.04000
Sinic	1783.64	940.82	0.01000
United States	770.21	445.87	0.00714

Table 5. Equilibrium T under linear and saturating PI-driven inflow.

5.2 Historical counterfactual

Table 6 compares the equilibrium that would have emerged if the transition rates estimated for the early career window (2000-2010) or the late window (2011-2016) had persisted indefinitely. The late window is shorter and its rates are estimated from younger cohorts, so the comparison should be read as a sensitivity exercise rather than a forecast. The largest negative margin changes are observed for the Anglosphere ex-US, Sinic and Other Western groups, while the Islamic, Japanese and Other Civilizations groups would see larger safety margins under late-window rates. These divergent patterns show that global AI/ML mobility is not moving in a single direction; different civilisations are on different trajectories, and a uniform policy response would ignore this heterogeneity. Because the late cohort is younger, the late-window equilibrium is likely biased downward for groups where career progression has not yet run its course. Even so, the exercise is useful because it shows that the current regime is not the only possible one: had the early-window rates persisted, the Anglosphere ex-US and Sinic groups would now have larger active pools, while the Islamic and Other Civilizations groups would have smaller ones. This asymmetry is what makes counterfactual policy analysis necessary.

Group	**T early**	**T late**	**delta_T**	**ΔT (%)**	**M_early**	**M_late**	**Margin early**	**Margin late**	**Δ margin**
Anglosphere ex-US	699.2	250.1	-449.16	-64.2	119.78	119.78	579.5	130.3	-449.2
Continental Europe	1117.0	791.3	-325.69	-29.2	129.07	129.07	987.9	662.2	-325.7
Hindu	902.2	608.4	-293.74	-32.6	66.04	66.04	836.1	542.4	-293.7
Islamic	185.5	1106.5	920.94	496.4	126.52	126.52	59.0	979.9	920.9
Japanese	184.3	235.3	51.04	27.7	50.48	50.48	133.8	184.8	51.0
Other Civilizations	164.4	673.8	509.35	309.8	104.60	104.60	59.8	569.2	509.3
Other Western	304.0	239.0	-65.05	-21.4	57.17	57.17	246.9	181.8	-65.0
Sinic	1392.6	936.4	-456.17	-32.8	112.34	112.34	1280.3	824.1	-456.2
United States	770.7	547.3	-223.32	-29.0	132.49	132.49	638.2	414.8	-223.3

Table 6. Historical counterfactual: equilibrium active pool and safety margin under early versus late transition-rate regimes.

Figure 3 shows the change in safety margin between the early and late transition-rate regimes.

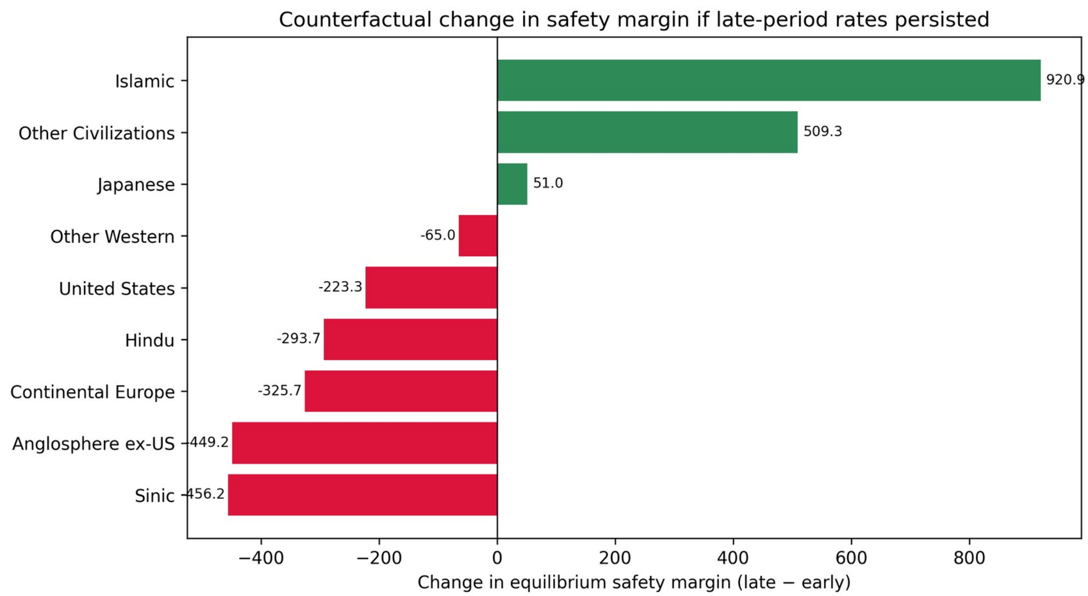

Figure 3. Change in safety margin between early and late transition-rate regimes. Positive values mean the late-window rates would produce a larger safety margin if they persisted.

5.3 Policy counterfactuals

Table 7 reports the single mechanical counterfactual with the largest margin gain per 10% lever change for each group. Reducing dropout is the dominant positive lever for every civilisation, which is consistent with the elasticity results in Table 3. The gain per 10% proportional reduction in d ranges from about 26 additional active researchers for the Japanese group to about 203 for the Sinic group, reflecting differences in cohort size and baseline attrition. No other single lever comes close to dropout reduction in terms of simulated margin gain per unit proportional change, although combinations of levers may be more efficient for some groups. The results also imply that policy need not focus on blocking early-career outflow. Reducing attrition among researchers who remain in the domestic system is a more efficient way to protect the active pool than preventing researchers from going abroad, because a researcher abroad is still in the global AI/ML system and may return. For the smallest groups, increasing the exogenous entry rate or improving the promotion of hit researchers to PIs can add additional margin, but dropout reduction remains the first-order model-implied target.

Group	**Lever**	**Direction**	**Change (%)**	**Margin gain**	**Gain per 10%**

Anglosphere ex-US	d	decrease	-10	88.9	88.9
Continental Europe	d	decrease	-10	141.0	141.0
Hindu	d	decrease	-10	168.3	168.3
Islamic	d	decrease	-10	101.2	101.2
Japanese	d	decrease	-10	25.8	25.8
Other Civilizations	d	decrease	-10	49.4	49.4
Other Western	d	decrease	-10	58.3	58.3
Sinic	d	decrease	-10	202.6	202.6
United States	d	decrease	-10	92.0	92.0

Table 7. Top positive mechanical counterfactual per group, measured by margin gain per 10% proportional lever change.

5.4 Uncertainty

Table 8 reports bootstrap 95% confidence intervals for the equilibrium active pool T and the domestic PI pool P_D. The intervals are wide, reflecting the small cohort sample and the extrapolation from individual careers to long-run steady states. For some groups the upper bound is an order of magnitude larger than the lower bound, indicating that the equilibrium is sensitive to resampling variation in the transition rates. This uncertainty should be interpreted as a warning against over-interpreting point estimates and as a reason to view the point-of-no-return distances as indicative rather than precise thresholds. Despite the width, the lower bounds for most groups remain above the minimum viable threshold, which supports the qualitative conclusion that all groups are currently above the point of no return. For the smallest groups the lower bound is closer to M, reinforcing the need for continued monitoring and for policy buffers.

Group	**T median**	**T 95% CI**	**P_D mean**	**P_D 95% CI**
Anglosphere ex-US	752	[260, 1595]	823	[207, 1517]
Continental Europe	1214	[761, 5174]	1532	[638, 5072]
Hindu	1497	[1492, 1502]	1421	[1366, 1446]

Islamic	892	[883, 899]	813	[691, 851]
Japanese	206	[96, 470]	173	[46, 402]
Other Civilizations	425	[189, 895]	486	[144, 848]
Other Western	514	[506, 518]	424	[356, 473]
Sinic	1785	[842, 3659]	2210	[756, 3586]
United States	775	[412, 2521]	831	[334, 2438]

Table 8. Bootstrap 95% confidence intervals for equilibrium T and domestic PI pool P_D.

Figure 4 displays the bootstrap intervals graphically.

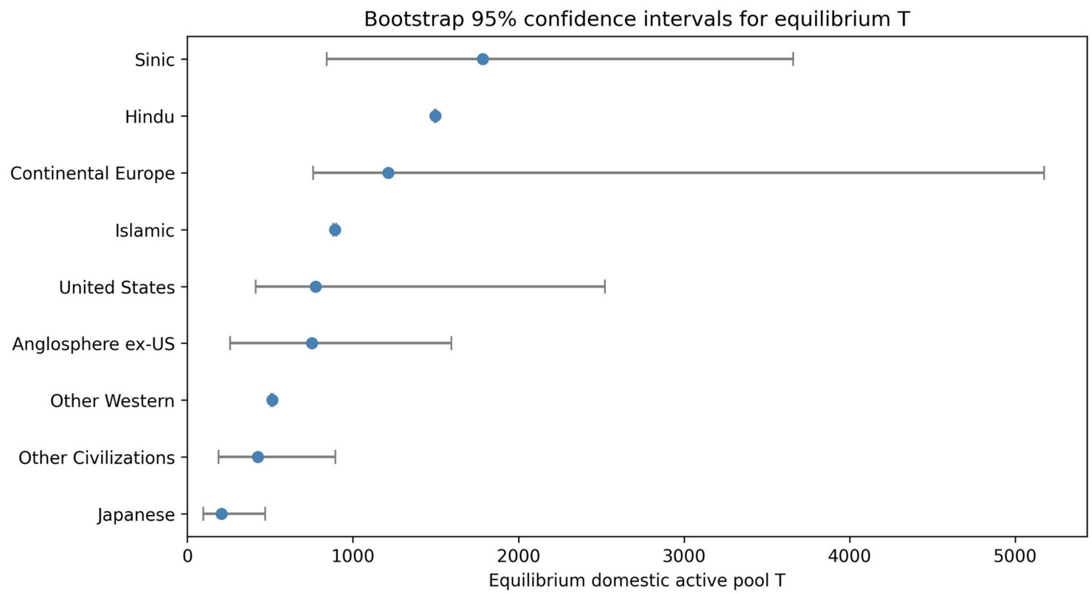

Figure 4. Bootstrap 95% confidence intervals for equilibrium T by group.

5.5 Synthesis

Taken together, the results provide a consistent picture. Exogenous entry and dropout are the two rates that most strongly determine the long-run viability of an AI/ML research community. Communities that are large in absolute terms are not necessarily safe if their coauthor intensity is high; conversely, small communities can be robust if their attrition is low and their recruitment pipeline is stable. The historical counterfactual shows that the current regime is not preordained: a shift in transition rates at the start of the AI boom would have produced different steady states for different civilisations. This is precisely why the framework is useful: it identifies which rate in which community is closest to a threshold, allowing policy to intervene before rather than after a collapse. The policy message is therefore both diagnostic and preventative. By tracking transition rates rather than net flows, policymakers can see which civilisation is approaching a point of no return and which lever offers the largest safety margin per unit of effort.

6. Discussion

The results support a transition-rate view of research policy. Rather than asking which country has a net inflow or outflow of researchers, the model asks which rate must be altered to keep a community above its minimum viable coauthor pool. This shift in focus has implications for how we conceptualise brain drain, design science and technology policy, and interpret civilisational diversity in AI/ML.

6.1 From net flows to transition rates

Most empirical studies of researcher mobility measure net flows, stocks or collaboration counts[12]. These indicators are useful for describing patterns, but they do not reveal the mechanisms that sustain or undermine a research community. A country may have a positive net inflow while simultaneously losing its domestic PI base through retirement or emigration, or it may have negative net flow but a healthy pipeline of new entrants. The transition-rate framework disaggregates these processes and shows that the same net flow can correspond to very different vulnerability profiles. For example, a high early-career outflow rate is less damaging than a high dropout rate because researchers abroad may return; a high dropout rate removes researchers from the system entirely. This distinction is lost in net-flow accounting but is central to point-of-no-return analysis.

First, exogenous entry (I0) is the closest point of no return for the active researcher pool in every group. A large proportional reduction in baseline recruitment would drive most communities to their threshold before mobility rates such as return or promotion became binding. This is consistent with the observation that AI/ML fields depend on a continuous pipeline of new graduate students and junior researchers[2]. Policies that sustain that pipeline, such as doctoral funding, visa routes for early-career researchers, and stable junior positions, are therefore first-order defences against a point of no return.

Second, among the mobility transition rates, dropout is the dominant lever. Its elasticity is near -2 for every group, and a simulated 10% reduction yields the largest margin gain per unit proportional change in the policy counterfactuals. Attrition matters because it removes researchers from every compartment, not just one. A 10% proportional reduction in dropout expands the safety margin more than comparably sized increases in return, hit generation or promotion. For groups with small safety margins, such as Japan, even modest attrition reductions may substantially delay the threshold. These counterfactuals are mechanical perturbations of the fitted rates; they identify the most sensitive transition levers, not the causal effect of any specific policy programme.

Third, the positive transition levers are not symmetric. PI promotion (p_D) and domestic hit generation (h_D) have positive but smaller elasticities than dropout reduction. Return from abroad (β) is also positive, though its effect is generally smaller than keeping researchers from leaving in the first place. The implication for policy is that retention is usually cheaper and more effective than return, but a balanced portfolio is still needed: a community without domestic PI growth cannot reproduce itself through attrition reduction alone.

Fourth, the historical counterfactual shows that the late-window rates, if they persisted, would change equilibrium margins in both directions. Several large groups would see lower safety margins, while Japan and some smaller groups would see higher margins. This heterogeneity cautions against treating AI/ML mobility as a single global trend. It also confirms that the model can detect temporal changes in transition rates, which is the prerequisite for the early intervention the framework is designed to support.

The transition levers also interact in ways that a single-rate elasticity cannot fully capture. For example, reducing dropout and increasing PI promotion together are likely to have a larger effect than the sum of the two individual perturbations, because more researchers survive to become PIs and those PIs then train additional early-career researchers through the endogenous inflow channel. Conversely, a simultaneous fall in exogenous entry and a rise in dropout can push a community to its threshold faster than either change alone. The model's steady-state and one-at-a-time counterfactuals are therefore a starting point; they identify the most sensitive margins but do not exhaust the policy design space.

The connection to civilisational diversity is direct. Each group's safety margin can be monitored over time, and interventions can be adjusted before the margin disappears. Because the model uses a fixed safety factor of 0.5 for the endogenous inflow parameter r, the policy recommendations are deliberately conservative: they do not push the system toward instability. That bounded approach is consistent with the goal of preserving diversity rather than maximising any single country's share.

It is important to stress that the counterfactuals reported in Tables 3 and 7 are mechanical perturbations of the fitted transition rates, not causal estimates of specific programmes. They identify which rates the model treats as most sensitive, and therefore where empirical policy evaluation is most urgent, but they do not by themselves show that a given intervention would achieve the simulated change.

6.2 Civilisational diversity as an innovation input

A second implication concerns the normative status of civilisational diversity. We treat diversity as an input to innovation rather than as a distributional afterthought[14]. A monocentric or tight-oligopoly structure in AI/ML may produce short-run efficiency gains through scale and agglomeration, but it also raises the risk of methodological lock-in, selection bias in training data, and reduced error correction. Recent work on multi-university teams shows that geographically dispersed collaborations can retain high impact, which suggests that distributing capacity across civilisations need not sacrifice quality[13]. By quantifying the safety margin for each research community, the framework makes it possible to argue for support of smaller communities on positive, innovation-systems grounds. Preserving multiple centres of AI/ML research is not a matter of slowing the frontier; it is a matter of ensuring that the frontier is not defined by a single set of institutions, languages, or problems.

6.3 Policy implications and early warning

The policy implications can be read as an early-warning architecture. A single dashboard that tracks the fitted transition rates, their bootstrap uncertainty, and the distance to M for each civilisation would allow policymakers to detect divergence before a community enters an irreversible decline. Interventions can then be calibrated to maintain a minimum safety margin rather than to maximise any one stock. This is the operational meaning of early intervention: not a forecast that a particular collapse will occur, but a structured way to keep the system away from a point of no return. It also frames high-skilled mobility as a strategic competition among jurisdictions for talent[15], in which the central question is not only who wins the current round but whether the global system retains enough diversity for future rounds[16]. If the dt of policy response is short enough, the model can be updated annually and divergence caught early, before any single civilisation approaches a point of no return. This is the mechanism through which technology monopoly, hegemonic concentration and oligopoly dead-ends can be avoided: by keeping every major research community above its minimum viable coauthor pool, the framework sustains the competitive diversity that underpins long-run technological progress. The framework is therefore not a prediction that a particular civilisation will collapse. It is a tool for ensuring that no single civilisation reaches a point where its collapse becomes self-sustaining, and that the global AI/ML system retains the diversity required for continued innovation.

6.4 Limitations

Several limitations should be acknowledged. OpenAlex affiliation and country assignments are noisy, especially for researchers with multiple affiliations. The civilisation grouping is a coarse aggregation; within-group heterogeneity is substantial. The model is a steady-state ODE and does not capture short-term dynamics, cross-civilisation spillovers, or the non-linear effects of network externalities. The cohort sample is small; the absolute equilibrium numbers should be interpreted as model-implied stocks rather than as census counts. Finally, the point-of-no-return threshold is a sufficient condition for collapse, not a necessary one: a community may decline for reasons outside the model even if T remains above M.

From a security-studies perspective, the framework is intentionally non-adversarial. It does not model deliberate recruitment campaigns, technology transfer, or strategic denial. Instead, it treats mobility as an aggregate transition process and asks when a community becomes unable to reproduce itself. That baseline is useful because it shows where defensive, capacity-building policies can be most efficient, but it does not replace classified or diplomatic assessments of technology competition. Future work could add a strategic layer by distinguishing between civilian and defence-relevant AI/ML pipelines, or by modelling the effects of targeted recruitment on specific subfields.

7. Conclusion

We have proposed and implemented a transition-rate framework for assessing how close AI/ML research communities are to a point of no return. The model converts OpenAlex publication records into civilisation-specific transition rates and solves for the equilibrium active researcher pool. All groups remain above their minimum viable coauthor threshold in the fitted model, but the distance to that threshold varies by an order of magnitude and is most sensitive to exogenous entry and dropout. Dropout is the dominant negative lever, and a simulated reduction is the single most efficient model-implied response for every civilisation. However, the closest point of no return is exogenous entry for most groups, which means that policies which sustain the pipeline of new researchers are first-order defences. The historical counterfactual and the bootstrap intervals remind us that the future is not determined by current rates; transition rates can change, and policy can be directed at the most fragile lever before a collapse.

The broader implication is that preserving civilisational diversity in AI/ML is compatible with, and may reinforce, scientific progress. A single dominant region or a tight oligopoly may achieve short-run scale economies, but it also risks methodological lock-in and reduces the set of problems that receive sustained attention. By monitoring transition rates and safety margins, policymakers can detect divergence early and intervene in a safety-factor-bound way. This is the practical meaning of the aspiration to avoid technology monopoly and oligopoly dead ends: not a prediction that any one civilisation will dominate, but a structured method for keeping the global system away from points of no return. Early, proportionate interventions that reduce attrition and sustain new recruitment can widen safety margins and preserve civilisational diversity in AI/ML.

7.1 Future work

Several extensions are natural. First, the model can be applied to other security-relevant fields such as semiconductor physics, quantum computing, biotechnology and energy materials. Second, the civilisation partition can be refined to a country or institution level, allowing bilateral migration flows and network externalities to be incorporated. Third, the ODE can be solved dynamically rather than at steady state, making it possible to forecast the time to threshold under alternative policy scenarios. Fourth, the minimum viable coauthor threshold can be made endogenous by modelling coauthorship as a matching process. Finally, the framework can be integrated with policy cost data to produce cost-effectiveness comparisons of alternative interventions.

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